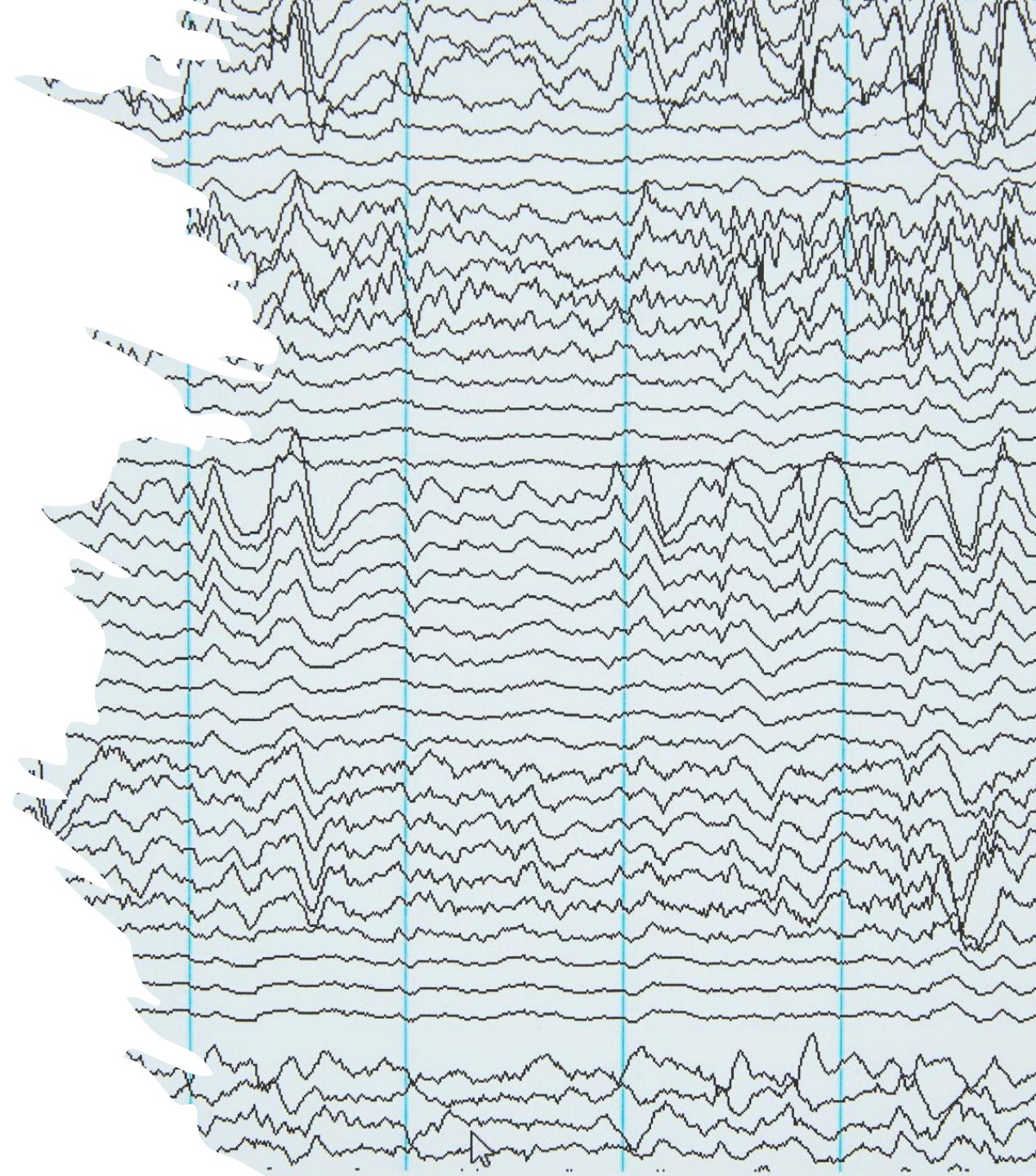


Hackathon Module: EEG Text-Based Reasoning



Hackathon Overview

- Project-based competition
- Teams build language-based reasoning systems to perform on raw EEG data
- Alzheimer's-focused tasks
 - We will not do **classification** instead conduct text-based reasoning like clinicians

Hackathon Day Overview

What you are competing on

- A working system (not a single model)
- Clinician-style reasoning in language (claims + evidence)
- Handling uncertainty and confounds (artifacts, drowsiness, noise)
- Clear recommended next step(s)

What you will receive:

- Raw EEG data
- Task prompts that require reasoning (not just classification)
- An evaluation based on a set of test cases or challenges that we will provide

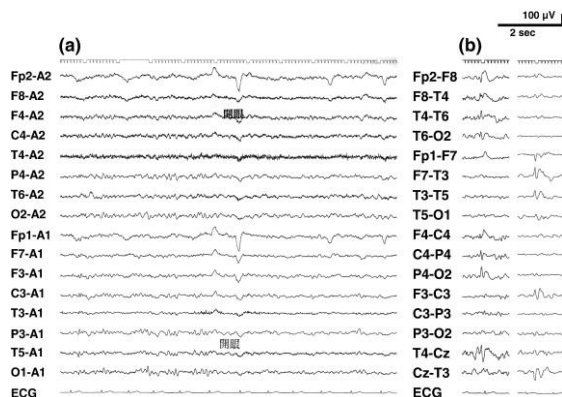
Clinicians Reason (They Don't Just Classify)

Clinical reasoning behaviors your system must emulate

- Your AI system must analyze raw EEG data then Generate and compare multiple clinical hypotheses
 - Extract signal features from raw EEG
 - Map features → interpretable patterns
 - Generate differential diagnoses
 - Compare hypotheses using evidence weighting
 - Output structured clinical reasoning text
- Explain evidence-for and evidence-against each hypothesis
- Track uncertainty (confidence + what would change your mind)

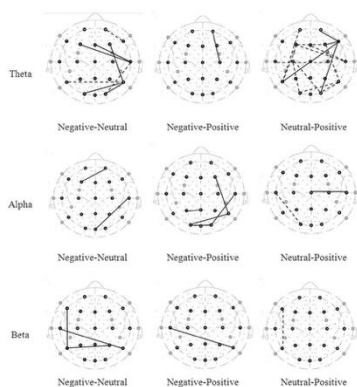
Stage 1 — Raw EEG → Biomarker Extraction

1: Input Biomarkers (Example Patient)



- 19-channel resting-state EEG
- Eyes closed
- 10–20 system

2: LLM Biomarker Extraction of EEG



Spectral Features

Posterior dominant rhythm: 6.5 Hz (↓ from age norm ~9–10 Hz)

Theta power: elevated (+35%)

Alpha power: reduced (−28%)

Connectivity

Posterior alpha coherence reduced

Complexity

Lempel-Ziv complexity ↓

Multiscale entropy ↓

```
PDR_slowed = true
theta_increased = true
alpha_reduced = true
posterior_coherence_reduced = true
complexity_reduced = true
epileptiform_activity = false
```


Stage 2 —Text-Based Hypothesis Generation

Hypothesis 1: Normal Aging

- **Supporting EEG Evidence**
 - ✓ Mild slowing can occur with age
- **Contradicting Evidence**
 - ✓ Theta elevation significant
 - ✓ Coherence reduction beyond age norm
 - ✓ Complexity reduction pronounced
- **Preliminary Likelihood: Low–Moderate**

Hypothesis 2: Mild Cognitive Impairment (MCI)

- **Supporting EEG Evidence**
 - ✓ Alpha reduction
 - ✓ Mild theta increase
 - ✓ Preserved reactivity
 - ✓ No epileptiform activity
- **Contradicting Evidence**
 - ✓ Network disconnection somewhat strong
- **Preliminary Likelihood: Moderate**

Hypothesis 3: Early Neurodegenerative Process (e.g., Alzheimer-type)

- **Supporting EEG Evidence**
 - ✓ Posterior slowing
 - ✓ Alpha→theta spectral shift
 - ✓ Reduced posterior coherence
 - ✓ Reduced complexity
- **Contradicting Evidence**
 - ✓ No severe global slowing
- **Preliminary Likelihood: Moderate–High**

Example Text-Based Reasoning Output

EEG demonstrates posterior dominant rhythm slowing with increased theta power and reduced alpha coherence. Signal complexity is reduced.

These findings exceed what is typically expected in normal aging and are more consistent with early neurodegenerative change.

Mild cognitive impairment remains plausible.

Multimodal confirmation recommended.

Data-Driven Layer (Objective)

Raw EEG → quantifiable biomarkers

Reasoning Layer (Transparent)

Biomarkers → hypothesis generation → evidence comparison

This makes the system:

- Interpretable
- Auditable
- Clinically defensible
- Suitable for regulatory framing

Why not Classical Pipelines?

This is how EEG analysis has been done for decades.

Pipeline Raw EEG → Filtering → Artifact rejection → FFT / Wavelet transform → Compute band powers → Connectivity (coherence/PLI) → Entropy metrics

Examples

PDR frequency → peak of power spectrum (FFT)

Theta power → band integration (4–8 Hz)

Coherence → magnitude-squared coherence

Lempel-Ziv complexity → deterministic algorithm

Characteristics

- Transparent
- Deterministic
- Clinically accepted
- Explainable
- Easy to validate

Limitations:

- Feature set is predefined
- Cannot discover novel biomarkers
- Limited nonlinear representation

AI-Based Feature Extraction (Learned Representations)

Here, biomarkers are not hand-crafted — they are **learned from data**.

Pipeline

Raw EEG → CNN / Transformer / GNN → Latent embedding space → Learned discriminative features
Latent features → mapped to interpretable biomarkers

GPU Access Request

GPU Access Request:

<https://ci.its.hawaii.edu/portal/extra/class-reg/spring-2026/s2026ics491b/>

Only one member per group should fill out the form for the whole group.

Please take care of this ASAP.